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Abstract

The vigorous development of artificial intelligence (AI) represented by large language models (LLMs) has rapidly promoted the updating and development of educational technology. Agentic workflows (AWs) built based on LLMs can realize complex tasks in the field of education, which allows the emergence of swarm intelligence (SI) through multi-agent collaboration^[1]. This study introduces the Agent4EDU (agent for education) framework, which outlines 4 application models in education from the two dimensions of degree of agency and degree of interaction, including human-AI collaboration, AI assistant, instruction execution, and general type. The proposed Agent4EDU framework discusses the paradigm of educational applications of AI agents and promotes the development of the field of AI for education.

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CCS Concepts

• **Applied computing** → Education; Interactive learning environments; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms.

Keywords

AI for education, agentic workflow, multi-agent system, AI agent, large language model

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1 Introduction

Large language models (LLMs) have been widely used in natural language processing, computer vision and other fields due to their powerful generative capabilities. Many researchers have also applied LLMs to the field of education, exploring the application of generative artificial intelligence in learning resource generation, Socratic dialogue and other aspects. Equation 1) outlines the general LLM calling method, which takes human thoughts and specific needs as inputs and uses the Transformer included in the LLM to process these inputs to obtain the generated output. By integrating

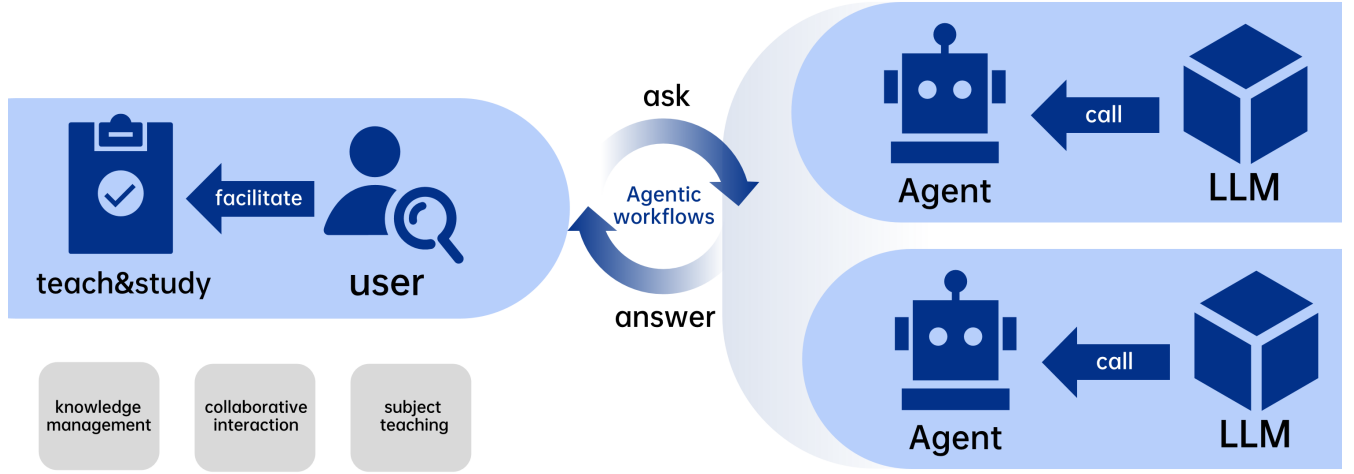


Figure 1: Framework Diagram of Agentic Workflows

users' ideas and needs, LLM can effectively respond to users.

$$\begin{cases} Output_{LLM} = \text{Transformer}(Input) \\ \text{s.t.} Input = \text{Human}(ideas, needs_{edu}) \end{cases} \quad (1)$$

LLMs-based agents have the ability to learn, make decisions, and collaborate autonomously, which enables them to handle complex educational tasks and explore potential development paths for future education in large-scale personalized learning. Equation 2) shows the process of a learner handling educational tasks by interacting with an LLM-based agent, where the learner mainly gives feedback on the agent's output. By providing ideas and educational needs to the autonomous agent, the learner and agent can continuously improve the final output together in the process of "AI-based questioning and learning" with the agent. This workflow shows how agents and learners can collaborate in the field of education.

$$\begin{cases} Output_{Teach\&Study} = \text{Feedback}(User, Agent(LLM(Input))) \\ \text{s.t.} Input = \text{Human}(ideas, needs_{edu}) \end{cases} \quad (2)$$

In order to formally discuss the application mode of LLM-based agents in education, we proposed the Agent4EDU framework, which aims to improve educational practice by using an agentic workflow (AW) based on multi-agent collaboration. Figure 1 shows the framework diagram of AW, in which learners ask questions, and then the multi-agent system (MAS) based on LLM obtains answers through AW. In the cycle of questions and answers, AW can enable learners to better improve educational practice. This study mainly focuses on exploring the development trend and application status of LLM and agents in education, and explores the application scenarios of AI for education based on the Agent4EDU education application framework. This study will provide new ideas and methods for the application of AI for education, and promote technological innovation and paradigm change in the field of education.

2 Paradigm Shift: Multi-agent systems drive AI change in education

2.1 Emergence of Swarm Intelligence: the Transition from Single-Agent to Multi-Agent System

Swarm intelligence (SI) originates from the study of social animal behavior. Individuals in these populations can generate powerful SI to deal with real-world problems through simple interaction rules of self-organization [2]. In MAS, multi-agent collaboration can also promote the emergence of SI [3]. Here, we formally describe the educational application of MAS using Equation 3), where n is the number of agents in MAS. These agents are coupled through AW and call LLMs to form and improve the final output. Multi-agent collaboration can effectively deal with complex problems in educational scenarios, which exceeds the task processing capabilities of single-agent. The SI that emerges in MAS enables it to better meet the needs of learners.

$$\begin{cases} Output_{Teach\&Study} = \text{Feedback}(User, \sum_{i=1}^n Agent_i) \\ (LLM(\text{Assign_Function}(Input, i))) \\ \text{s.t.} Input = \text{Human}(ideas, needs_{edu}) \end{cases} \quad (3)$$

Agent, as a concept in the field of computer science, is defined as an entity that can perceive its environment and make decisions on its own. It is characterized by autonomy and flexibility and has been widely used in education. Specifically, in terms of learning methods, Bruner [4] and Ramadhevi [5] used agents to promote collaborative learning, which was achieved through agent-based simulation. Agents are also used to form the core of distance education framework in order to provide personalized courses and learning experiences[6]. In terms of training applications, agent-based frameworks are also used to establish security training and corporate education to promote knowledge transfer [7-8]. Finally, in terms of educational support tools, Stummer[9], Walters [10] and

Winkler [11] have also developed agent-based models and technologies that can adjust teaching tasks and improve students' learning experience.

As technology develops, learners' educational needs are becoming more segmented and specific, which makes the complexity of educational tasks grow rapidly. Therefore, unlike single agents, MAS has also begun to be widely used in the field of education due to its stronger task processing capabilities in real scenarios. For example, the research of Xie [12], Nazir [13] and Molnar [14] revealed the great potential of MAS in improving teaching efficiency and optimizing resource allocation. Furthermore, the work of Boudabous [15], Singh [16] and Miura [17] further expanded the application of MAS in the field of education and promoted the development of AI for education.

2.2 Transition of Paradigm: From Preset Models to LLM-Driven Evolution

LLM-based MAS is designed to perceive the environment and make precise decisions based on perception to maximize its performance in real-world tasks. By using LLM as the core of systems for educational purposes, these systems are able to perceive, reason, and make decisions and actions autonomously. LLM provides direct processing of natural language to efficiently solve educational tasks. This promotes the change of paradigm from preset models to LLM-driven evolution. For example, online writing platforms may have only been able to make revision suggestions based on semantics and polishing rules, but now they can use LLM-based MAS to generate more specific revision suggestions, thereby effectively helping learners improve their writing skills.

At present, the research and development of agent-based education systems mainly focuses on the education platforms development and learning analytics. In terms of education platform development, Khasianov [18] developed a multi-agent education platform that uses agents to enhance learners' initiative and thus speed up their knowledge acquisition efficiency. In order to further improve the quality of teaching, Yan [19] developed an agent-based computer network-assisted teaching platform. In terms of learning analytics, Chen [20] and Simon [21] introduced agents to analyze public health and higher education systems, thereby providing traceable data support for policy making and long-term development. Hantono [22] and Bruner [23] discussed the potential of augmented reality and agent networks in improving learning efficiency and motivation.

LLM can process complex natural language input and dynamically adjust responses based on context and user input, which makes LLM-based educational systems more practical. By integrating LLM-based agents in educational systems, those systems' language processing and flexible interaction capabilities are greatly enhanced. For example, Zheng et al. [24] used an LLM-supported embedded agent for language understanding and sub-goal planning, and constructed semantic graphs through visual observation to obtain symbolic representations. MathBoard proposed by Jiang et al. [25] is based on LLM and gradually generates board writing (BW) by integrating a multi-agent collaborative framework, thereby effectively promoting learners' knowledge construction.

2.3 Knowledge Customization: Transitioning from General Domains to Subject-Specific Teaching

At present, agent-based systems have been widely used in education, medicine, finance and other fields. In the field of information retrieval, Coneglian [26] used agents to effectively extract information to promote the efficiency and accuracy of information retrieval. In the field of control engineering, Ma [27]'s research on multi-agent collaboration integrated the efficient task processing capabilities of agents into the control engineering classroom to improve the educational model. In the field of education, researchers such as Yang [28], Yonah [29], Alexandru [30], and Jawahar [31] used AI agents as core components of teaching platforms, providing learners with real-time feedback and personalized support to optimize the learning process and promote interaction. Furthermore, Gu [32]'s work on higher education simulation further verified the effectiveness of agents in educational system analysis.

Focusing on the application of agents in the field of education, they can provide technical support for teaching scenarios such as personalized learning, automatic question and answer, and Socratic guidance. Their application scope has gradually shifted from general knowledge fields to more targeted subject teaching. For example, Qi [33] proposed the IntelliChain framework, an educational application framework for enhanced Socratic dialogue with LLMs and knowledge graphs. Xie [34] and Yan [35] focused on physical education, using a multi-agent teaching management system and a computer network-assisted teaching platform to promote the all-round development of students by improving teaching efficiency. Samra [36] applied data mining agents and intelligent simulation technology to clinical skills education, improving clinical teaching effectiveness by optimizing the learning environment. Dubovi [37] used an interpretable agent-based model (ABM) to reveal the specific advantages of biological learning by making micro-mechanisms visible. Although agent technology has brought more flexible and personalized learning experiences to education and effectively improved the efficiency of teaching and learning [38], the ethical issues of these educational technologies must be considered to ensure the credibility and sustainable development of educational technologies [39-40].

3 Agent4EDU supported Educational Application Framework

The application of AI agents in the field of education has produced many specific cases, in which agents are often used to play roles such as instructors and study partners to promote learners' knowledge construction process and provide timely emotional support to learners. To better analyze the application types of agents in education, we divide the educational applications of Agent4EDU into four levels according to the autonomy of agents and the depth of human-AI interaction, as shown in Figure 2, where the interaction level is divided into general type, AI assistant, instruction executor and human-AI collaboration type from shallow to deep; the autonomy level is divided into general type, instruction executor, human-AI collaboration and AI assistant from low to high [41]. Overall, the interaction depth and autonomy level of the

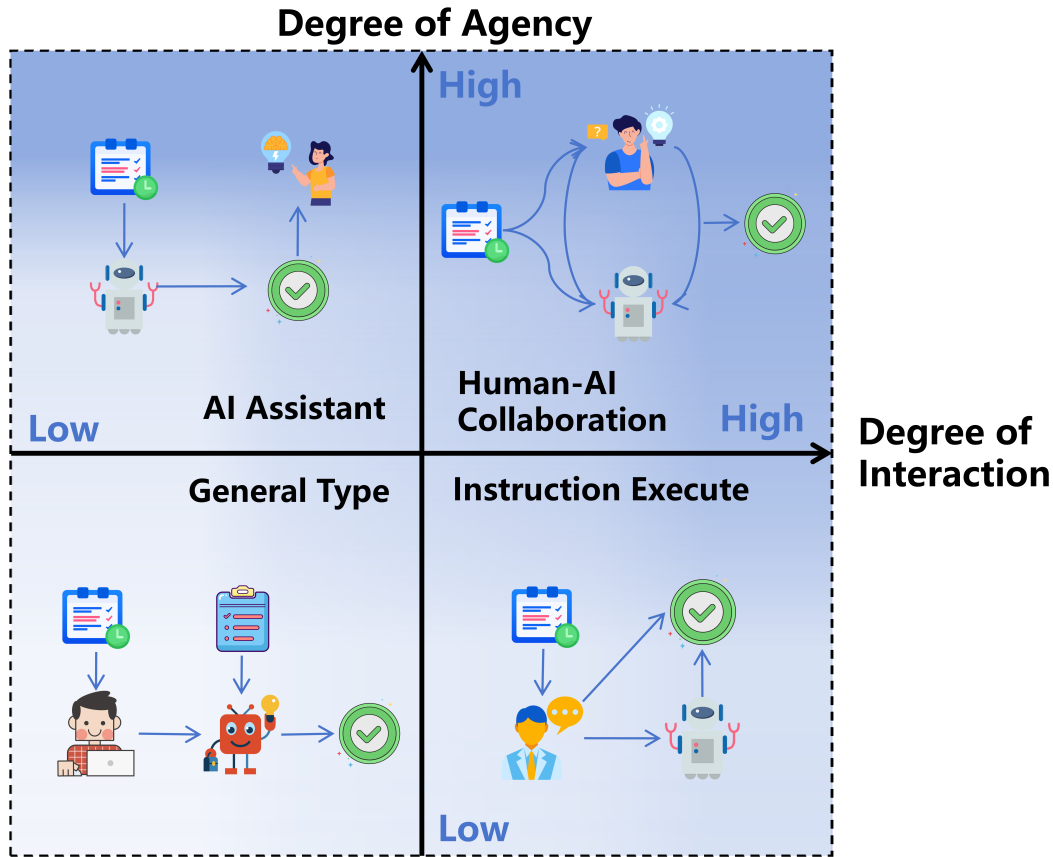


Figure 2: Agent4EDU Educational Application Framework

Agent4EDU framework show a progressive relationship, reflecting the differences in human-AI interaction modes in education. We believe that the proposed Agent4EDU framework will provide forward-looking guidance for the educational application of AI agents to promote the development of AI for education.

3.1 Human-AI Collaboration

Human-AI collaboration is defined as the highest level of interaction in the Agent4EDU framework. In this type, the AI represented by agents works closely with users, such as collaborating on a painting or art creation. In this mode, agents do not just respond to commands or provide basic learning support; instead, agents collaborate and co-create with users in depth. The main task of agents is to identify learners' learning needs and goals, interact through the user interface to obtain user input and feedback, and improve output based on this information to advance the improvement of educational tasks. The human-AI collaboration type is suitable for educational scenarios such as virtual laboratories and collaborative learning, which enhances the interactivity of learning, thereby potentially increasing learners' enthusiasm and participation.

3.2 AI Assistant

The AI assistant type is at the highest level of autonomy, where agents can proactively make decisions and provide effective learning support to learners. By analyzing learners' learning data and behavior information, the AI assistant played by the agent can quickly understand the user's needs and feedback, and create personalized learning suggestions and learning resource suggestions based on the user's preferences. When the AI assistant receives input or requests from learners, it can use its knowledge database and algorithms to analyze the input, generate suggestions in combination with historical learning data, and wait for further feedback or requests until the learner makes a decision. This interactive process will continue until the problem is resolved. By providing precise and effective support, agents can bring a personalized learning experience to learners.

3.3 Instruction Execution

In the instruction execution mode, agents receive commands from learners and perform tasks as instructed. This makes the system functionality easy to implement. These agent-based educational applications can meet different learner needs and make the learning experience more personalized. The AW begins when the agent receives an explicit command from the learner. The agent looks

at the command to find the required action. It performs tasks such as searching for information or tutoring learners to complete homework. This mode is well suited for automated learning tasks such as vocabulary memorization and standardized testing, but it is weakly coupled with learners at the collaborative level.

3.4 General Type

General-type agents form the foundational layer. These agents provide basic educational resources and services, such as online courses and learning databases, and are able to perform routine tasks and simple requests. In this mode, agents can handle simple queries and interactions, and usually involve one-way information transfer. These agents have low autonomy. When learners' requests are sent to agents, agents will call predefined functions or services to respond. The solutions it provides do not have complex processing and decision-making processes. This type of agent is suitable for providing basic learning resources, answering common questions, or providing course and grade information.

4 Conclusion

In this research, we review the efforts and achievements of agents in applying AI to education. It also introduces the Agent4EDU framework. By exploring the integration of large language models and agent technology, AWEs provide new solutions for personalized and adaptive learning, improve students' learning efficiency and experience, reduce teachers' workload, and promote continuous innovation in educational technology.

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References

- [1] Shi J, Zhao J, Wu X, Xu R, Jiang Y-H, He L. 2025. Mitigating reasoning hallucination through multi-agent collaborative filtering. *Expert Systems with Applications*, 263(2025), 125723. <https://doi.org/10.1016/j.eswa.2024.125723>
- [2] Chen Z-W, Tang K, Jiang, Y-H, Duan, B, Zhao C, Meng, C. 2025. MU-PTO: Multi-unmanned aerial vehicle supported passive tracking optimization for dynamic maritime lost items. *Proceedings of the 2024 International Conference on Industrial Automation and Robotics*, 24–29. New York, NY, USA: Association for Computing Machinery. <https://doi.org/10.1145/3707402.3707407>
- [3] Jiang Y-H, Shi J, Tu Y, Zhou Y, Zhang W, Wei Y. 2024. For learners: AI agent is all you need. In Wei Y, Qi C, Jiang Y-H, Dai L, editors, *Enhancing educational practices: strategies for assessing and improving learning outcomes*, 21–46. Nova Science Publishers. <https://doi.org/10.52305/RUIG5131>
- [4] Bruner L. 2023. A multi-agent modeling social network analysis of cooperative learning groups within a simulated adult education classroom learning environment. Accessed: Mar. 22, 2024. [Online]. Available: https://tigerprints.clemson.edu/all_dissertations/3285
- [5] Ramadhevi M, Rajaprakash S. 2022. An overview on applications and developments in education on agent based ontological cooperative learning. *Journal of Pharmaceutical Negative Results*, 723–728.
- [6] Zaytsev E, Khalabiya R, Stepanova I, Bunina L. 2021. Multi-agent system in virtual education. In *INTED2021 Proceedings, IATED*, 4501–4506.
- [7] Babalola A, Cheung C, Manu P, Yunusa-Kaltungo A. 2023. Development of a framework integrating agent-based modelling, building information modelling and immersive technologies for construction training and education. In *Book of Proceedings*, 471.
- [8] Yang J-S. 2022. Dynamics of firm's investment in education and training: an agent-based approach. *Comput Econ*, 60(4): 1317–1351. <https://doi.org/10.1007/s10614-021-10206-6>
- [9] Stummer C, Kiesling E. 2021. An agent-based market simulation for enriching innovation management education. *Cent Eur J Oper Res*, 29(1): 143–161. <https://doi.org/10.1007/s10100-020-00716-3>
- [10] Walters G. 2019. Evaluating conversation agent impact on student experience in a distance education course. University of South Florida. Accessed: Mar. 22, 2024. [Online]. Available: <https://search.proquest.com/openview/44b51fedd0ba26605e949da4fb469515>
- [11] Winkler R, Hobert S, Salovaara A, Söllner M, Leimeister JM. 2020. Sara, the lecturer: improving learning in online education with a scaffolding-based conversational agent. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, Honolulu HI USA: ACM, 1–14. <https://doi.org/10.1145/3313831.3376781>
- [12] Xie S, Xu J. 2023. Design and implementation of physical education teaching management system based on multi-agent model. *Int J Comput Intell Syst*, 16(1): 172. <https://doi.org/10.1007/s44196-023-00349-9>
- [13] Nazir M, Noraziah A, Rahmah M. 2023. Students' performance prediction in higher education using multi-agent framework-based distributed data mining approach: a review. *International Journal of Virtual and Personal Learning Environments (IJVPLE)*, 13(1): 1–19.
- [14] Molnar A. 2023. Improved quality of online education using prioritized multi-agent reinforcement learning for video traffic scheduling. *IEEE Transactions on Broadcasting*. Accessed: Mar. 22, 2024. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/10073590/>
- [15] Boudabous S, Kazar O, Laouar MR, Eom SB. 2021. A multi-agent system for ubiquitous learning for hospital education. *International Journal of Information Systems in the Service Sector (IJSSS)*, 13(3): 1–17.
- [16] Singh AK, Nag A, Karforma S, Mukhopadhyay S. 2019. Implementation of multi-agent based digital rights management system for distance education (DRMSDE) using JADE. *International Journal of Advanced Computer Science and Applications*, 10(3).
- [17] Miura M. 2015. Project based learning with multi-agent simulation in liberal arts education. In *2015 3rd International Conference on Applied Computing and Information Technology/2nd International Conference on Computational Science and Intelligence, IEEE*, 242–247.
- [18] Khasianov A, Suleymanov D, Marchenko A. 2017. Three agent platform approach for digital education environment. In *INTED2017 Proceedings, IATED*, 8580–8587.
- [19] Yan P. 2016. The research on the building of university physical education model based on the construction of computer multi-agent. *Revista Ibérica de Sistemas e Tecnologias de Informação*, 18A: 270.
- [20] Chen H-J, Xue H, Kumanyika S, Wang Y. 2017. School beverage environment and children's energy expenditure associated with physical education class: an agent-based model simulation. *Pediatric Obesity*, 12(3): 203–212.
- [21] Simon N. 2016. Agent based modeling of development of Russian Federation higher education system. In *INTED2016 Proceedings, IATED*, 6869–6877.
- [22] Hanton BS, Nugroho LE, Santosa PI. 2016. Review of augmented reality agent in education. In *2016 6th International Annual Engineering Seminar (InAES)*, IEEE, 150–153.
- [23] Bruner L. 2023. A multi-agent modeling social network analysis of cooperative learning groups within a simulated adult education classroom learning environment. Accessed: Mar. 22, 2024. Available: https://tigerprints.clemson.edu/all_dissertations/3285/
- [24] Zheng K, Zhou K, Gu J, Fan Y, Wang J, Di Z, Wang XE. 2022. Jarvis: A neuro-symbolic commonsense reasoning framework for conversational embodied agents. *arXiv preprint arXiv:2208.13266*.
- [25] Jiang Y-H, Li R, Wei Y, Jia R, Shao X, Hu H, Jiang B. 2024. Synchronizing verbal responses and board writing for multimodal math instruction with LLMs. In *The 4th Workshop on Mathematical Reasoning and AI at NeurIPS'24*, Vancouver, Canada, 46–59, [https://openreview.net/forum?id=\\$esb1rV8N12](https://openreview.net/forum?id=$esb1rV8N12)
- [26] Coneglian CS, Fusco E, de Castro Botega L. 2014. Using ontological semantic agent to data extraction in the context of big data: a tool applied to higher education institutions. In *Proceedings of the International Conference on Semantic Web and Web Services (SWWS)*, The Steering Committee of The World Congress in Computer Science, Computer, 1–6.
- [27] Ma JH, Zhang H, He BF. 2014. Applied-information technology with multi-agent cooperation in education model control engineering. *Applied Mechanics and Materials*, 662: 255–258.
- [28] Yang LP, Xu DW, Wang W. 2014. The application of intelligent agent in network platform for undergraduate's innovation and entrepreneurship education. *Applied Mechanics and Materials*, 687: 2149–2152.
- [29] Yonah E. 2014. A multi-agent system to support ICT-based distance learning through modeling of learner needs: the case of Bachelor of Education at University of Nairobi. PhD Thesis, University of Nairobi. Accessed: Mar. 22, 2024. Available: <http://erepository.uonbi.ac.ke/handle/11295/73675>

- [30] Alexandru A, Tirziu E, Tudora E, Bica O, “Enhanced education by using intelligent agents in multi-agent adaptive e-learning systems,” *Studies in Informatics and Control*, vol. 24, no. 1, pp. 13–22, 2015.
- [31] Jawahar J, “A study on role of intelligent agent in education,” *International Journal of Applied Engineering Research*, vol. 10, no. 76, pp. 497–501, 2015.
- [32] Gu X, Blackmore KL. 2015. A systematic review of agent-based modelling and simulation applications in the higher education domain. *Higher Education Research & Development*, 34(5): 883–898. <https://doi.org/10.1080/07294360.2015.1011088>
- [33] Qi C, Jia L, Wei Y, Jiang Y-H, Gu X. 2024. IntelliChain: An integrated framework for enhanced Socratic method dialogue with LLMs and knowledge graphs. In *Conference Proceedings of the 28th Global Chinese Conference on Computers in Education (GCCCE 2024)*. Chongqing, China, Global Chinese Conference on Computers in Education, 116–121.
- [34] Xie S, Xu J. 2023. Design and implementation of physical education teaching management system based on multi-agent model. *Int J Comput Intell Syst*, 16(1): 172. <https://doi.org/10.1007/s44196-023-00349-9>
- [35] Yan P. 2016. The research on the building of university physical education model based on the construction of computer multi-agent. *Revista Ibérica de Sistemas e Tecnologias de Informação*, 18A: 270.
- [36] Samra HE, Li AS, Soh B, Alzain MA. 2016. A conceptual model for an intelligent simulation-based learning management system using a data mining agent in clinical skills education. In *2016 4th International Conference on Enterprise Systems (ES)*, IEEE, 81–88.
- [37] Dubovi I, Lee V. 2018. Agent-based models to support bioscience learning in nursing education. *International Society of the Learning Sciences, Inc.[ISLS]*. Accessed: Mar. 22, 2024. Available: <https://repository.isls.org/handle/1/557>
- [38] Zhuang X, Wu H, Shen X, Yu P, Yi G, Chen X, Hu T, Chen Y, Ren Y, Zhang Y, Song Y, Liu B, Lan M. 2024. TOREE: Evaluating topic relevance of student essays for Chinese primary and middle school education. In *Findings of the Association for Computational Linguistics ACL 2024*, edited by Ku L-W, Martins A, Srikumar V, 5749–65. Bangkok, Thailand and virtual meeting: Association for Computational Linguistics.
- [39] Chan A, Salganik R, Markelius A, Pang C, Rajkumar N, Krashennnikov D, Langosco L, He Z, Duan Y, Carroll M. 2023. Harms from Increasingly Agentic Algorithmic Systems. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, 651–666. <https://doi.org/10.1145/3593013.3594033>
- [40] Wang, W, Tu, Y, Song, L, Zheng, J, Wang, T. 2021. An Adaptive Design for Item Parameter Online Estimation and Q-Matrix Online Calibration in CD-CAT. *Frontiers in Psychology*, 12. <https://doi.org/10.3389/fpsyg.2021.710497>
- [41] Jiang Y-H, Liu T-Y, Zhuang X, Hu H, Li R, Jia R. 2024. Enhancing educational practices with multi-agent systems: A review. In *Wei Y, Qi C, Jiang Y-H, Dai L, editors, Enhancing Educational Practices: Strategies for Assessing and Improving Learning Outcomes*, 47–65. Nova Science Publishers. <https://doi.org/10.52305/RUIG5131>